

SURAJ MISHRA

Software Engineer | Backend, AI/LLM & Platform Engineering

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PROFESSIONAL SUMMARY

- Software Engineer with nearly 4 years of experience designing and delivering scalable travel-tech and Industrial IoT platforms, with hands-on production experience building AI/LLM-powered systems.
- Currently leads a six-member engineering team at Yatra Online Limited, owning architecture and delivery across B2C and B2B travel platforms, including train, flight, bus, cab, SEO/AEO, AI-powered CMS, and analytics systems.
- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline for semantic search — embeddings generation, vector indexing in Milvus and OpenSearch, and retrieval-orchestrated response generation via Ollama — powering the SEO/AEO semantic search experience.
- Experienced in Java, Spring Boot, Spring WebFlux, Next.js, distributed systems, event-driven architecture, caching, API integration, platform modernization, and production delivery.
- Strong grounding in system design and design patterns — microservices decomposition, event-driven and adapter-based integrations, circuit breaker/retry resilience patterns, caching strategies, and idempotency — applied consistently across booking, search, and content platforms.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript

Backend & Web: Spring Boot, Spring WebFlux, Spring Cloud, Spring Data JPA, Spring Security, Spring AOP, Hibernate, Next.js, REST APIs, Microservices, B2C/B2B platforms

Data & Messaging: MongoDB, PostgreSQL, Redis, Aerospike, Apache Kafka, Kafka Connect, Apache Spark, MQTT

Cloud & Tools: AWS, Azure IoT Hub, Docker, Git, Gradle, Jira, Keycloak, Okta

AI & Search: LLM integration (Ollama), Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases (Milvus), OpenSearch, Semantic Search, Claude, AI-assisted development, SEO, AEO, AI-powered content workflows

Design Patterns & System Design: Microservices, Event-Driven Architecture, Circuit Breaker, Retry/Backoff, Idempotency, Singleton, Factory, Strategy, Observer, Adapter, CQRS-style read/write separation, Caching Strategies, Load Balancing, Rate Limiting, High Availability, Fault Tolerance, Horizontal Scalability

PROFESSIONAL EXPERIENCE

Software Engineer / Team Lead | Yatra Online Limited

June 2025 – Present

- Lead a six-member engineering team and own end-to-end delivery — from business discovery and architecture through implementation, testing, release, and production support.
- Partner directly with business stakeholders to translate operational problems into scalable technical solutions; drive planning, task allocation, code reviews, release coordination, and delivery across multiple travel products.
- Design shared services and API contracts for B2C customer journeys and B2B partner integrations, enabling channel-specific experiences while reusing common search, pricing, booking, payment, and post-booking capabilities.

Train Booking Platform Modernization

- Architected and delivered a new train-booking engine to replace constraints in the legacy Java platform; identified bottlenecks, tightly coupled modules, unreliable integrations, and performance limitations before defining the target architecture.
- Stabilized the existing Java stack while building the new engine in parallel, using backward-compatible APIs and phased integration to preserve business continuity and reduce migration risk.
- Designed reusable services for search, availability, fares, passenger details, booking, payment, and post-booking workflows with validation, structured exception handling, timeouts, controlled retries, fallbacks, and correlation-based logging.
- Used Redis and Aerospike for low-latency search/session access and Kafka for asynchronous booking, payment, notification, and analytics events, improving scalability and service decoupling.

Flight Search API Platform

- Designed an API wrapper over Yatra's existing Flight SRP data, shielding consumers from complex legacy and supplier-specific structures through stable, normalized, frontend-friendly contracts.
- Implemented request validation, DTO transformation, error mapping, timeout/retry controls, and caching for frequently searched routes, enabling reuse of schedule, pricing, availability, and route data across product and SEO experiences.

- Created reusable flight-search contracts for both B2C and B2B consumers, reducing direct dependency on legacy Flight SRP response structures.

Bus and Cab Booking Systems

- Enhanced end-to-end cab workflows covering location search, supplier availability, pricing, vehicle selection, booking, cancellation, and status tracking; introduced a common adapter model to normalize supplier responses.
- Improved bus workflows across route search, operator listing, live seat availability, fare revalidation, and booking; revalidated seats and fares before confirmation to reduce stale-inventory failures.
- Applied idempotency and transaction-state validation to prevent duplicate bookings during retries, using short-lived caching, Kafka events, supplier-level timeouts, and structured monitoring to isolate downstream failures.

SEO, AEO and AI-Powered CMS

- Architected and deployed a scalable Java and Next.js SEO platform designed to handle approximately 100,000 concurrent users, supported by distributed caching, database indexing, stateless services, and asynchronous processing.
- Built SEO and Answer Engine Optimization capabilities using LLMs and created an AI-powered CMS enabling business users to generate, review, manage, and publish search-optimized content.
- Designed and built an end-to-end RAG (Retrieval-Augmented Generation) pipeline for semantic content search: generated and indexed embeddings in Milvus and OpenSearch, and orchestrated retrieval-grounded LLM responses via Ollama, improving content discoverability and relevance for AEO.
- Implemented real-time publishing and Kafka-driven cache invalidation so approved CMS changes propagate immediately across relevant pages, using MongoDB/PostgreSQL for persistence and Redis/Aerospike for high-throughput reads.

Analytics and System Behaviour Engine

- Designed an event-driven analytics engine to analyze application and user behavior across train, flight, cab, bus, SEO, and booking services.
- Used Kafka-based event collection and correlation IDs to track search volume, conversion, latency, failures, payment issues, and user drop-off, helping engineering and business teams make data-driven decisions.

Software Engineer | IoT83

June 2023 – June 2025

- Developed scalable backend services for Flex83, an end-to-end Industrial IoT platform that helps OEMs build and deploy custom IIoT applications while retaining control and long-term ownership.

Identity and Access Management

- Built a reusable IAM service using Keycloak and Okta to simplify authentication and authorization across applications; improved API resilience with retry policies, timeouts, concurrency handling, and structured error management.

ETL Transformation Service

- Designed a configurable ETL application supporting multiple sources and sinks with runtime settings; integrated it with a Java and Apache Spark processing engine for scalable data transformation and provisioning.

Connector Service Architecture

- Decoupled the Connector Service from Kafka Connect by introducing a Connector Agent, enabling dynamic scaling, flexible lifecycle management, and improved maintainability across multiple connectors.

EDUCATION

Bachelor of Technology in Computer Science | G.L.A. University

May 2019 – May 2023

- Built a strong foundation in operating systems, computer architecture, networking, security, software engineering, web technologies, and machine learning through academic and project work.

CORE STRENGTHS

- Technical leadership, team coordination, and stakeholder collaboration
- System design, distributed architecture, and project delivery
- Problem solving, performance optimization, and production troubleshooting
- Time management, adaptability, ownership, and continuous learning